higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

(8080175)

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09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 5 pages and a formula sheet of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

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QUESTION 1: AC THEORY

1.1 Draw a labelled circuit diagram of the following, and include their output waveforms:

1.1.1 RC differentiator (3)

1.1.2 RC integrator (3)

1.2 Given: $Z_1 = 5 + j\ 15\ \Omega$
$Z_2 = 10 - j\ 21\ \Omega$

In a parallel circuit Z_1 is connected across Z_2 with a supply voltage of 25 V.

1.2.1 Calculate the currents through each branch, including total current. (6)

1.2.2 Draw the current phasor diagram. (3)
[15]

QUESTION 2: POWER SUPPLY

2.1 Show by means of a neatly labelled block diagram the components of a power supply. (3)

2.2 An AC supply is stepped down, using a transformer at 50 Hz. It is connected to a rectifier with a capacitor filter. The capacitor has a value of 2200 µF, the load resistor is 220 Ω and the maximum voltage is 33,941 V.

If the rectifier is a bridge, calculate:

2.2.1 The output voltage (4)

2.2.2 The load current (2)

2.2.3 The ripple factor (2)

2.3 Briefly explain the operation principle of a full-wave voltage doubler and state ONE disadvantage thereof. (4)
[15]

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QUESTION 3: AMPLIFIER

3.1 Given information from the fixed bias amplifier.

h_{ie} = 13 kΩ	h_{fe} = 80	$h_{re} = 2{,}5\times10^{-4}$
h_{oe} = 35 µA / V / siemens	R_C = 2,4 kΩ	R_B = 300 kΩ

Calculate the following using an approximate method:

3.1.1 A_i (1)

3.1.2 A_v (2)

3.1.3 Z_{in} (2)

3.1.4 Z_{out} (2)

3.2 Draw the circuit diagram of a differential operational amplifier which uses TWO transistors. (5)

3.3 Calculate the value of the feedback resistor if the input voltage is 20 millivolts, output voltage is 120 millivolts and input resistor is 2 kΩ as was read from the non-inverting operational amplifier. (3)

[15]

QUESTION 4: IC AND TRANSDUCERS

4.1 Explain in brief how damage from electrostatic discharge (ESD) happens. (3)

4.2 List THREE advantages of integrated circuits. (3)

4.3 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (4.3.1–4.3.5) in the ANSWER BOOK.

4.3.1 In a resistance strain gauge, the resistance wire changes when subjected to elongation/compression due to internal stress.

4.3.2 The application of variable capacitance pressure gauges is displacement or pressure.

4.3.3 In the photomultiplier, the secondary electron emission will occur due to incident light.

4.3.4 In a thermocouple, an emf will be generated across the junction of the two dissimilar metals when the junction's temperature is stable.

4.3.5 Guage factor is the unit change in resistance per unit change in the dimensions.

(4 × 5) (5)

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4.4 The following circuit is constructed with information R_t = 10 kΩ @ 25 °C and the constant B = 3600.

Calculate the value of the R_t @ 30 °C. (4)

[15]

QUESTION 5: ELECTRONIC PHASE CONTROL

5.1 Briefly explain in words the difference between a semiconductor SCR and a TRIAC. (4)

5.2 Draw a neatly labelled block diagram of a closed loop system. (4)

5.3 List TWO applications of a closed loop system. (2)

[10]

QUESTION 6: MEASURING INSTRUMENTS

6.1 Explain the term *digital signal*. (2)

6.2 Briefly explain the operation principle of a staircase ramp digital voltmeter. (6)

6.3 Name TWO voltmeters that are applicable in measuring instruments. (2)

[10]

QUESTION 7: OSCILLATORS

7.1 Oscillators form an integral part of the electronic industry since various types of waveforms for different applications can be generated by such circuits.

List THREE factors that may affect the oscillation frequency. (3)

7.2 Describe fully the principal operation of the Schmitt-trigger circuit. (3)

7.3 Draw a neatly labelled circuit diagram of an equivalent circuit of a quartz crystal and show the output wave form of the crystal impedance opposite frequency. (4)

[10]

QUESTION 8: LIQUID CRYSTAL DISPLAY

8.1 Name FIVE advantages of a liquid crystal display. (5)

8.2 Draw a neatly labelled circuit diagram of a liquid crystal display. (5)

[10]

TOTAL: 100

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FORMULA SHEET

$$I = \frac{V}{R}$$

$$P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + \ldots = I_1R_1 + I_2R_2 + I_3R_3 + \ldots$$

$$I_T = I_1 + I_2 + I_3 + \ldots = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \ldots$$

$$T = RC$$

$$T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt}$$

$$V_C = \frac{1}{RC}\int v_i\,dt$$

$$X_L = 2\pi fL$$

$$X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L$$

$$Z = R - jX_C$$

$$Z = R + j(X_L - X_C)$$

$$I_T = \frac{V_T}{Z_T}$$

$$V_R = I_TR$$

$$V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C)$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$\frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$BW = f_2 - f_1$$

$$I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2}$$

$$\frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$Z_T = \frac{R\,(jX_L)}{R + jX_L}$$

$$I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$I_T = I_R - jI_L$$

$$\frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C}$$

$$I_T = \frac{V}{R} + j\frac{V}{X_C}$$

$$I_T = I_R + jI_C$$

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$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$$I_T = I_R - j\left(I_L - I_C\right)$$

$$a + jb = \sqrt{a^2+b^2}\,/\tan^{-1}\frac{b}{a} = r\,\underline{/\theta}$$

$$r\,\underline{\angle\theta} = r\left(\cos\theta + j\sin\theta\right)$$

$$Q = \tan\theta$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}}\,V_m = 0{,}707\,V_m$$

$$V_{dc} = \frac{2}{\pi}\,V_m = 0{,}637\,V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}$$

$$V_{dc} = \frac{1}{\pi}\,V_m = 0{,}318\,V_m$$

$$PIV = V_m$$

$$PIV = 2\,V_m$$

$$R_{r(rms)} = 0{,}385\,V_m$$

$$V_{r(rms)} = 0{,}305\,V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}}$$

$$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2fC}$$

$$V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L}$$

$$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L} \cdot V_{dc}$$

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} \cdot V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R} \cdot V_{r(rms)}$$

$$r' = rX_C\left(\frac{R + R_L}{R \,.\, R_L}\right)$$

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$$V'_{dc} = V_{dc} - I_{dc}R_1$$

$$V'_{dc} = \frac{R_L}{R_L + R_1} \cdot V_{dc}$$

$$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$$

$$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$$

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1}$$

$$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i}$$

$$V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$$

$$I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} \cdot R_S$$

$$V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b}$$

$$C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10}$$

$$R_e = \frac{V_e}{I_e} \approx \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$R_{b2} = \frac{1}{10}\beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce}$$

$$i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

$$A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L}\right)\left(\frac{R_b T}{R_{bT} + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

$$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L}$$

$$Z_1 = h_{ie}$$

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$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe}h_{re}}{h_{ie} + R_s}} \qquad Z_2 = \frac{1}{h_{oe}}$$

$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i \qquad A_p = \frac{h_{fe^2} R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2 \qquad Z_{IN} = R_B // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1 \qquad I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe}\left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}}\right) \qquad A_i = \frac{I_0}{I_1}$$

For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.

$$Z_L = R_c // R_L \qquad I_1 = \frac{R_e I_i}{R_e + Z_1}$$

$$CMRR = \frac{A_{dm}}{A_{cm}} \qquad CMRR\ (dB) = 20log\frac{A_{dm}}{A_{cm}}$$

$$I_e = \frac{V_e}{R_e} \qquad I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C} \qquad g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left(\frac{R_f}{R_1}\right) . V_i \qquad V_0 = \left(\frac{R_f}{R_1} + 1\right) . V_i$$

$$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3\right) \qquad V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3) \qquad V_0 = -(I_1 + I_2 + I_3) R_f$$

$$V_0(t) = -\frac{1}{RC} \int V_i(t) \qquad V_0(t_b) = -\frac{1}{RC} \int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

$$t = \frac{1}{f} \qquad A_v = -\frac{R_s}{R_1}$$

$$R_2 = \frac{R_1 R_s}{R_1 + R_s} \qquad f_c = \frac{1}{2\pi R_s C}$$

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$$V_0(t) = -RC\frac{dV_i(t)}{dt}$$

$$A = \frac{R_f}{X_c}$$

$$A = -\frac{R_f}{R_s}$$

$$V_0(t) = -R_f C\frac{d}{dt} \cdot v_i \sin \omega t$$

$$t = R_f C$$

$$V_0 = A\left(V_r - V_i\right)$$

$$V_0 = \frac{R_f}{R_s}\left(V_2 - V_1\right)$$

$$V_0 = V_2 - V_1$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$$

$$L_T = L_1 + L_2 + 2M$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$$

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

$$f_0 = \frac{1,5}{RC}$$

$$f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$$

$$t_1 = 0,7\ R_2 C_1$$

$$t_2 = 0,7\ R_1 C_2$$

$$f_0 = \frac{1}{1,4\,RC}$$

$$V_i = I_{c2} R_e + V_{be(ON)}$$

$$f_0 = \frac{1,443}{\left(R_A + 2R_B\right)C}$$

$$t = 1,1\ RC$$

$$t_{low} = 0,693\left(R_B\right)C$$

$$t_{high} = 0,693\left(R_A + R_B\right)C$$

$$t_T = t_{low} + t_{high}$$

$$K = \frac{\Delta R/R}{\Delta l/l}$$

$$\sigma = \Delta l/l$$

$$R = \rho\frac{1}{\pi d^2/4}$$

$$\sigma = \frac{S}{E}$$

$$Resolution = \frac{1}{amount\ of\ turns}$$

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$$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$$

$$R_t = Ae^{B/T}$$

$$T = 273 + °C$$

$$V_A = \frac{R_2}{R_1 + R_2} \cdot V_T$$

$$V_B = \frac{R_t}{R_t + R_3} \cdot V_T$$

$$V_{AB} = V_A - V_B$$

$$A_v = \frac{V_0}{V_i}$$

$$V_{Hall} = kIH$$

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